# A Self-Supervised Attribution Method for Explaining Neural Networks

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# Motivation

**Attribution methods** Methods that produce an explanation by scoring input features.

Existing attribution methods:

- Require tuning method-specific hyperparameters.
- Need to perform multiple iterations or do backpropagation at inference time.

# Intuitive Idea

**Assumption** A good attribution map should only highlight elements that, if isolated from the input, allow the model to predict the same outcome.

- Pair the existing model with a separate module to produce attribution maps.
- The new module is trained to produce attribution maps that preserve the capabilities of the original model, when used as a weight of the input.

$$\min_{\mathbf{S}\in[0,1]^N} \|f(\mathbf{x}) - f(\mathbf{x} \otimes \mathbf{S})\| + \gamma \|\mathbf{S}\|$$

# Moving to Neural Networks

When working with neural networks, we can define the problem in terms of loss function:

$$\min_{\mathbf{S}\in[0,1]^N} \|f(\mathbf{x}) - f(\mathbf{x}\otimes\mathbf{S})\| + \gamma\,\|\mathbf{S}\|$$

$$\Downarrow$$

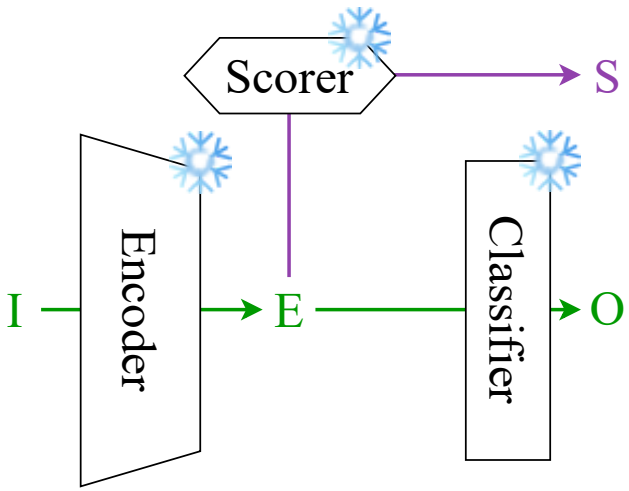
$$\min_{\mathbf{S}\in[0,1]^N} \mathcal{L}(\mathbf{x}\otimes\mathbf{S}; f) + \gamma\,\|\mathbf{S}\|$$

# Architecture

The scoring module works in the embedding space of the target model and produces the attribution map with a single forward pass.

To guide scores prediction, the head of the scorer is composed of:

- **ReLU** Cuts off negative scores (i.e., not relevant),
- **Sigmoid** Stabilizes scores into $[0.5, 1]$,
- **Rescale** Ensure the highest score is $1$ and lowest is $0$.

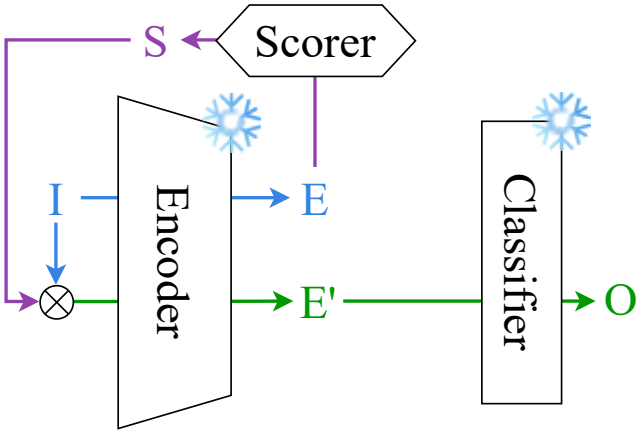


# Training Problem (1/2)

$$\mathcal{L}(\mathbf{x}, \mathbf{S}; f) = \mathcal{L}_{\text{task}}(\mathbf{x} \otimes \mathbf{S}; f)$$ Preserve original loss

$$+ \gamma_1 \left( \frac{1}{|\mathbf{S}|} \sum_i^{|S|} s_i \right)^2$$ Encourage sparsity

$$+ \gamma_2 \frac{1}{|\mathbf{S}|} \sum_i^{|S|} (s_i \cdot (1 - s_i))^2$$ Encourage binary-like scores



# Training Problem (2/2)

We can self-calibrate the hyperparameters $\gamma_1$ and $\gamma_2$ during training:

$$\gamma = \texttt{clip}\left\{\frac{1}{\mathcal{L}_{\text{task}}(\mathbf{x} \otimes \mathbf{S}; f) - \mathcal{L}_{\text{task}}(\mathbf{x}; f) + \varepsilon}, [0, 1]\right\}$$

**Intuitive idea** When the distance between the loss functions of the weighted and original input is:

- Large: the penalty terms are less important.
  $\Rightarrow$ preserve the original capabilities first.
- Small: reintroduce the penalties.
  $\Rightarrow$ can focus back to attribution maps.

# Modalities

Depending on the type of data, the scorer can be implemented as:

- **Text** Small transformer encoder.
- **Image** U-Net-style upsampling branch.
- **Multimodality** One scorer for each modality.

# Datasets

| Modality | Name | Task |
|---|---|---|
| Text | Tweet Sentiment | Sentiment analysis, short texts. |
| | IMDB | Sentiment analysis, long texts. |
| | LIAR | Fact checking. |
| Image | MNIST | Image classification, grayscale, low-resolution. |
| | CIFAR-10 | Image classification, colored, low-resolution. |
| | Imagenette | Image classification, colored, high-resolution. |
| Multimodal | Flickr8k | Image-caption alignment. |
| | Hateful Memes | Hateful content detection. |
| | SNLI-VE | Visual entailment. |

# Metrics

**Faithfulness** Is the explanation reflecting the underlying process?

- Average drop,
- Deletion AUC.

**Clarity** Is the explanation visually appealing?

- Complexity,
- Sparsity.

# Baselines

- Saliency,
- Guided Backpropagation,
- Integrated Gradients,
- DeepLIFT,
- DeepLIFT-SHAP,
- Gradient-SHAP.

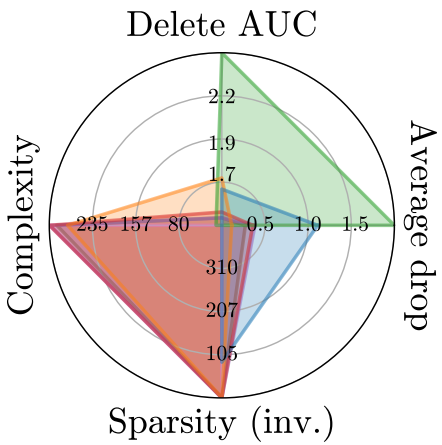
# Results

## Quantitative (1/4)

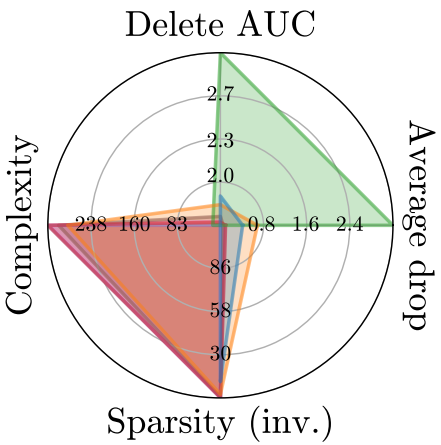
- Saliency is the method with the best Complexity and Sparsity results.
- The other baselines achieve the best results in terms of Average Drop and Deletion AUC.
- Our method balances faithfulness and clarity.

# Results

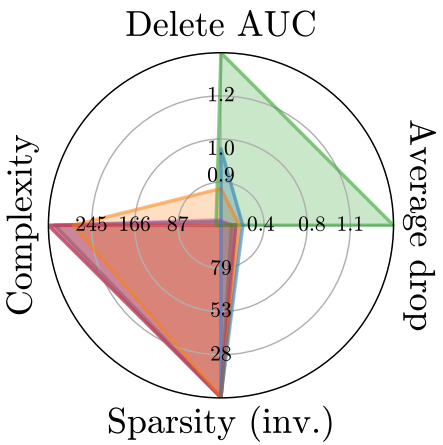
## Quantitative (2/4)




(a) Tweet Sentiment




(b) IMDB

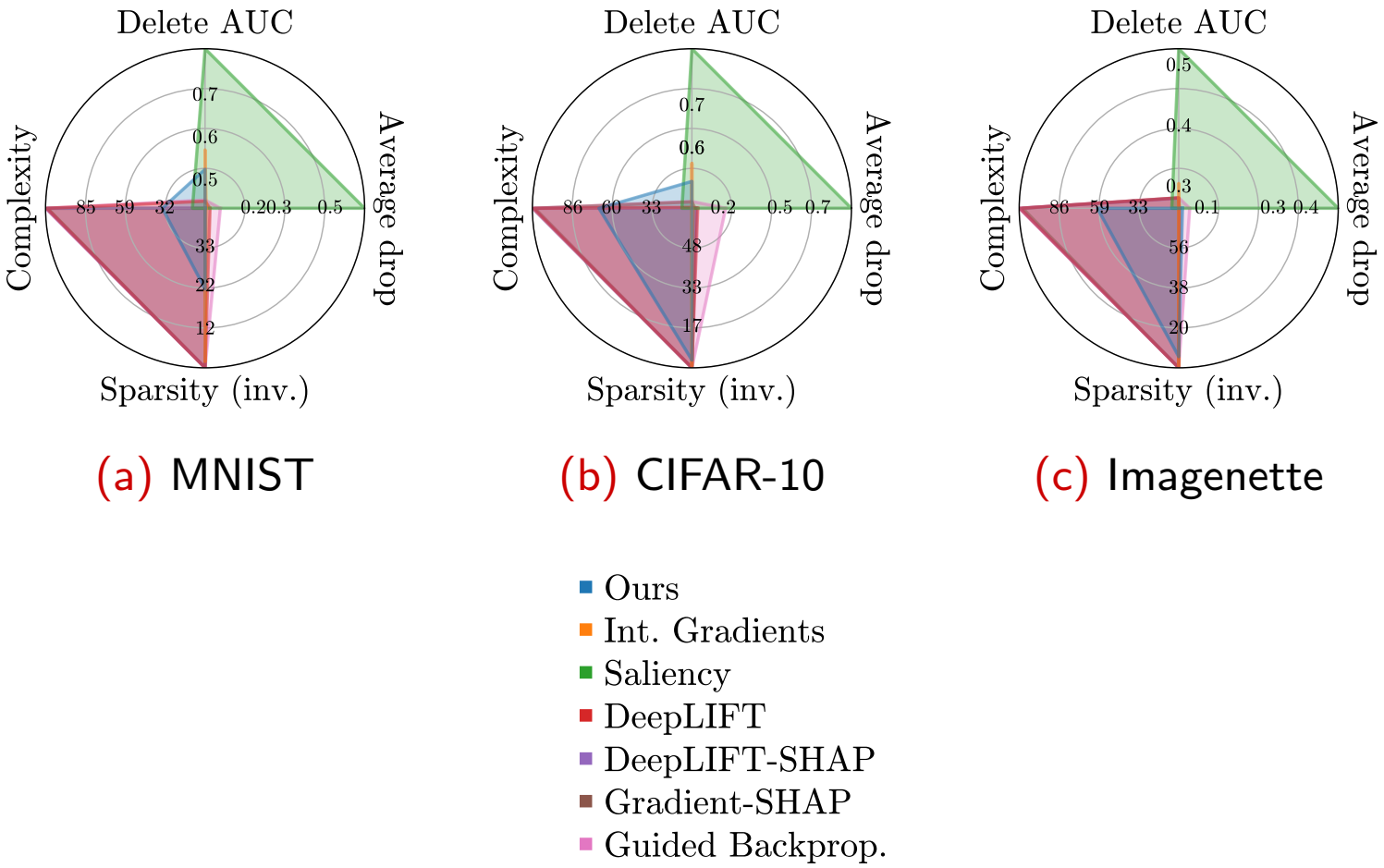



(c) LIAR

- Ours
- Int. Gradients
- Saliency
- DeepLIFT
- DeepLIFT-SHAP
- Gradient-SHAP
- Guided Backprop.

# Results

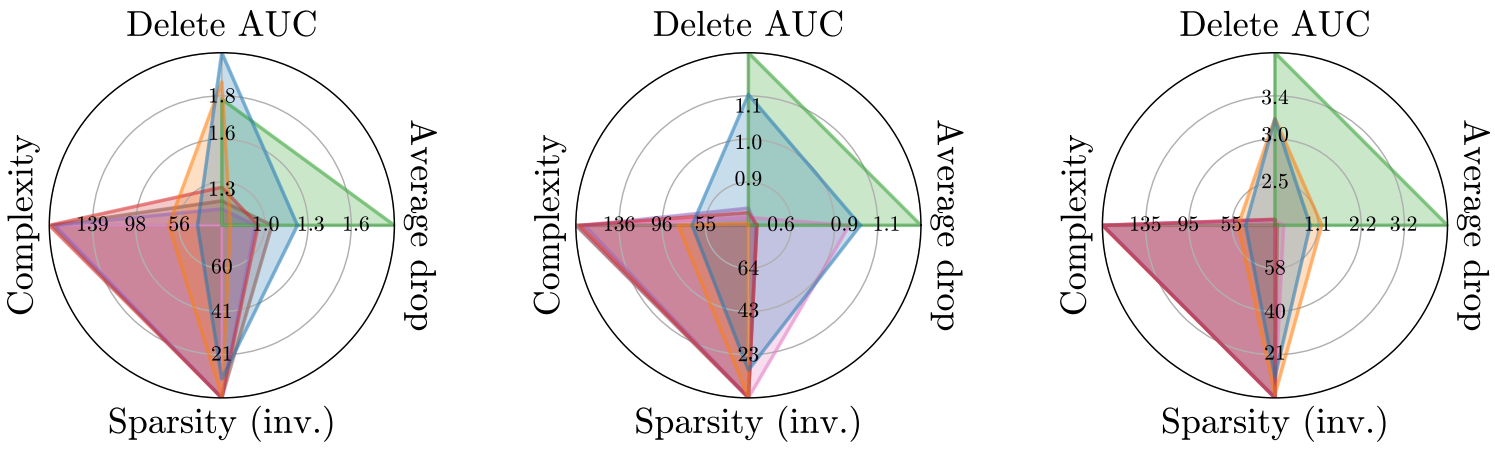
## Quantitative (3/4)




(a) MNIST (b) CIFAR-10 (c) Imagenette

# Results

## Quantitative (4/4)




(a) Flickr8k (b) Hateful Memes (c) SNLI-VE

- Ours
- Int. Gradients
- Saliency
- DeepLIFT
- DeepLIFT-SHAP
- Gradient-SHAP
- Guided Backprop.

# Results

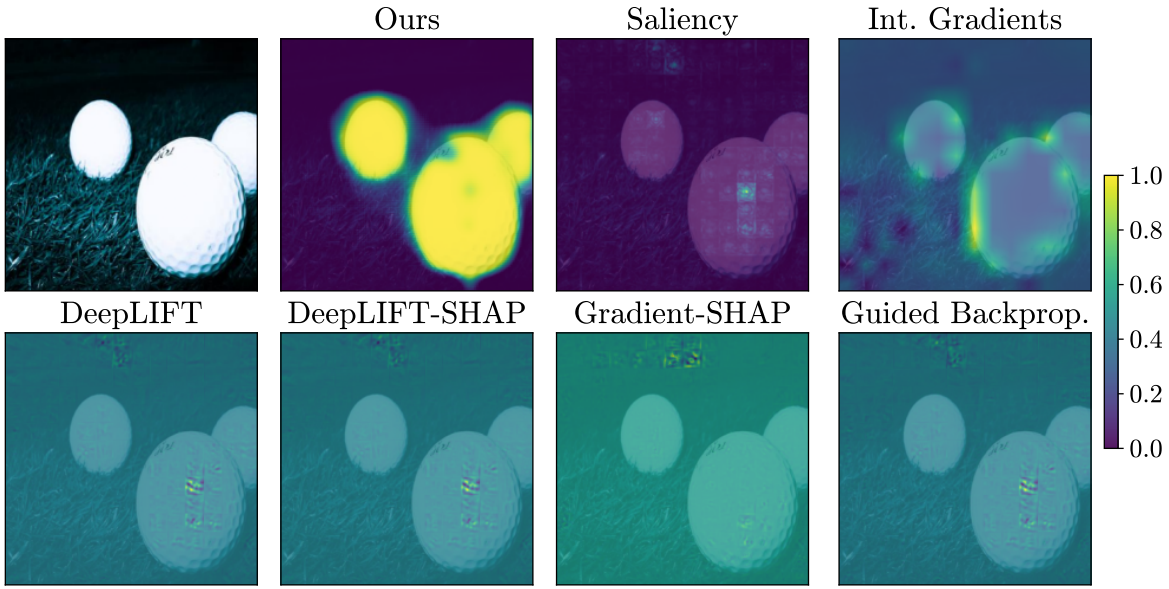
## Qualitative

- Saliency is the method that tends to assign high importance scores to a very small subset of the input space.
- The other baselines tend to give some degree of imporance to the whole input.
- Our method tend to only select a subset of the input as relevant.

# Results

## Qualitative

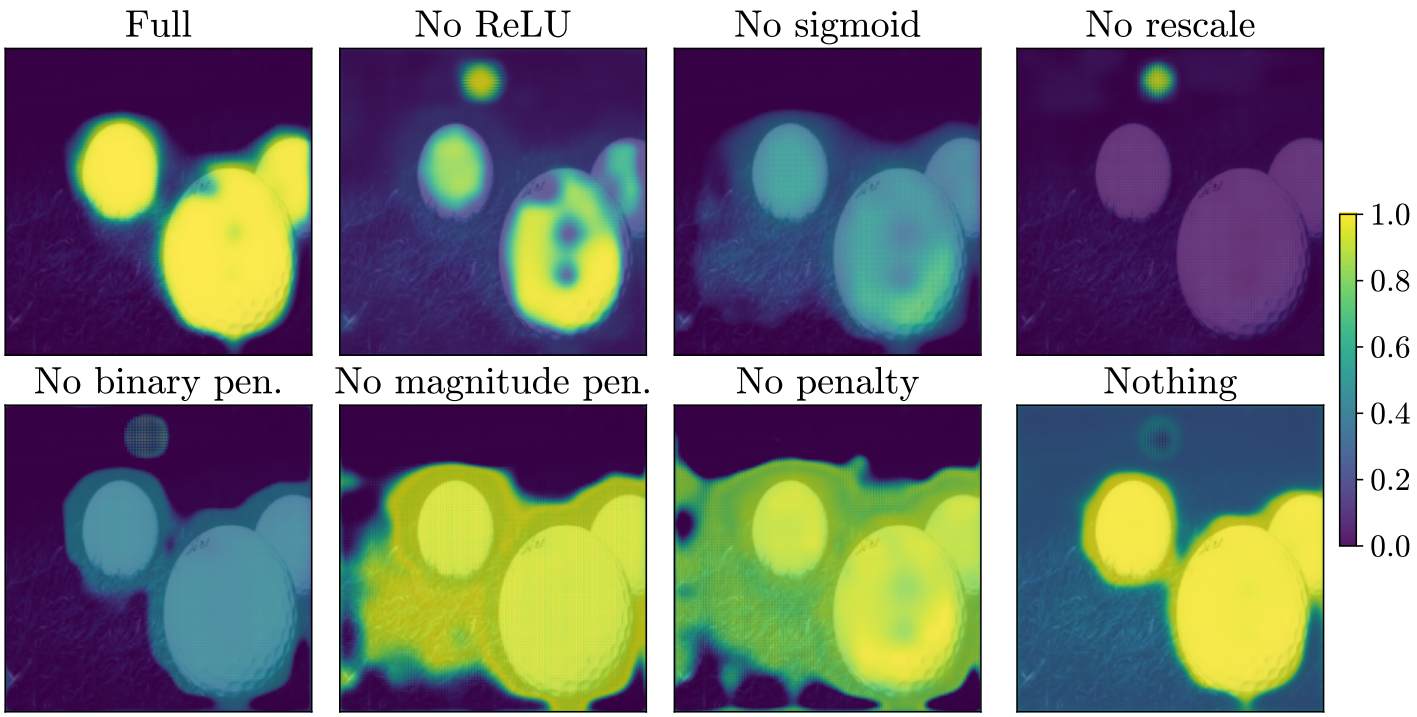
| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Ours** | This | is | a | great | movie. | Too | bad | it | is | not | available | on | home | video.</s> |
| **Saliency** | This | is | a | great | movie. | Too | bad | it | is | not | available | on | home | video.</s> |
| **Guided backprop** | This | is | a | great | movie. | Too | bad | it | is | not | available | on | home | video.</s> |
| **Int. Gradients** | This | is | a | great | movie. | Too | bad | it | is | not | available | on | home | video.</s> |
| **DeepLIFT** | This | is | a | great | movie. | Too | bad | it | is | not | available | on | home | video.</s> |
| **DeepLIFT-SHAP** | This | is | a | great | movie. | Too | bad | it | is | not | available | on | home | video.</s> |
| **Gradient-SHAP** | This | is | a | great | movie. | Too | bad | it | is | not | available | on | home | video.</s> |



# Results

## Ablation study

- Each design choice contributes to improve either faithfulness or clarity.



# Conclusion

- We proposed an attribution method that is trained in a self-supervised way, without introducing method-specific hyperparameters.
- We showed its applicability as a data-agnostic method for neural networks.
- We observed that among existing methods, ours has the best trade-off between model faithfulness and visual clarity both quantitatively and qualitatively.